

Seminar 1

1)

A	B	C	D	
$\{ \neq 0, \neq D \}$	$\{ \neq A, \neq D \}$	$\{ \neq A, \neq D \}$	$\{ 1, 3, 5, 7, 9 \}$	$\Rightarrow 8 \cdot 8 \cdot 7 \cdot 5 = 64 \cdot 35 = 2240$
8 options	8	7	5 options	

2) a) $2^4 = 3$

b) $2^9 = 512$

c) $\underbrace{\quad\quad\quad\quad\quad\quad\quad}_{2^9} \Rightarrow \begin{matrix} 1 \text{ if odd} \\ 0 \text{ if even} \end{matrix}$

d) $\underbrace{\quad\quad\quad\quad\quad\quad\quad}_{2^9} \Rightarrow \begin{matrix} 1 \text{ if even} \\ 0 \text{ if odd} \end{matrix}$

e) $\underbrace{\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad}_{2 \text{ ways}}$

L1, Ex 6: A B C D E F G H

$$\{C, H\} = \{H, C\}$$

6) $A = \{1, 2, \dots, 10\}$

a) $\{3\} \rightarrow \{4, 5, \dots, 10\} \quad 2^7 - 1 + 1 = 2^7 = 128$

b) $\boxed{1} \quad 2 \quad \boxed{3} \quad 4 \quad \boxed{5} \quad 6 \quad \boxed{7} \quad 8 \quad \boxed{9} \quad 10$

9 7 5 3 1

At each step, we choose the opponent of the smallest number with that opponent.

$$\{1, 3\}$$

$$\{2, 5\}$$

$$\{4, 7\}$$

$$\{6, 8\}$$

$$\{9, 10\}$$

$$9 \cdot 7 \cdot 5$$

$$9 \cdot 7 \cdot 5 \cdot 3 = 9!!$$

$$(2n \text{ people} \rightarrow n \text{ pairs}) = (2n-1)!!$$

$$8) \quad 4 = 1 + 0 + 0 + 0$$

$$4 = 3 + 1 + 0 + 0$$

$$4 = 2 + 1 + 1 + 0$$

$$4 = 2 + 2$$

$$4 = 1 + 3$$

$$4 = 1 + 2 + 1 + 0$$

$$4 = 1 + 1 + 2 + 0$$

$$4 = 1 + 1 + 1 + 1$$